FINAL REPORT - GROUP 6

G-46 (ESCOBOT)

Written by Gilbert Scott, Gregor Watt, Iain Harkness, Stewart Legge, Kieran Baird, and Joao Lopes

EFFORT BY EACH GROUP MEMBER FOR FINAL REPORT:

Gilbert, Iain, Gregor, Joao and Stewart 100%

Kieran 70%

Alex 10% (Website)

Course code and name:	B39VS Systems Project
Type of assessment:	**Group / Individual** *(delete as appropriate)*
Coursework Title:	Final Report
Student Name:	Gilbert Scott
Student ID Number:	H00346281

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Coursework Title:	Final Report
Student Name:	Iain Harkness
Student ID Number:	H00321201

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Type of assessment:	**Group** *(delete as appropriate)*
Coursework Title:	Final Report
Student Name:	Gregor Watt
Student ID Number:	H00341198

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Type of assessment:	**Group**
Coursework Title:	Final Report
Student Name:	Kieran Baird
Student ID Number:	H00345958

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Type of assessment:	**Group**
Coursework Title:	Final Report
Student Name:	João Lopes
Student ID Number:	H00342606

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Course code and name:	B39VS Systems Project
Type of assessment:	**Group**
Coursework Title:	Final Report
Student Name:	Stewart Legge
Student ID Number:	H00344138

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Student Signature *(type your name):* *Stewart Legge*

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ABSTRACT

As the need for robotics in all areas of life increases, one area where there is a demand for more automated systems is the healthcare sector. Among many other possible services, there has been a need identified for more electronic solutions for the secure transportation of medical supplies from department to department, to allow for the medical professionals to redirect their attention to other vital tasks. This paper is a report of one such device that was designed and developed and aims to detail how a sensor-based system may be implemented into use in the medical sector.

CHAPTER 1 INTRODUCTION

PROJECT OVERVIEW

We designed a solution for the problem of transporting medicine throughout a hospital with minimum human interaction in order to free up manpower and relieve pressure on the hard working staff. Our solution is in the form of a self driving robot.

AIMS AND OBJECTIVES

The aim of the project, as specified in the brief, is to create an automated system with the purpose of transporting medical supplies between different points in a predetermined path. This robot must also have a secure unit in which supplies can be safely stored, and a system to ensure that only authorized persons can access the unit.

COMPANY ROLES

B39VS Team Members Roles

		Company Name: Your company's name here			Assigned Roles								
SN	Student ID	Surname	Forename	email	MD	Sec	EW	DE	SE	PR	ME	WE	WD
1	H00321201	Harkness	Iain	ih2001@hw.ac.uk			x				x		
2	H00345958	Baird	Kieran	Kab2000@hw.ac.uk		x						x	
3			Alex							x			x
4	H00342606	Lopes	Joao		x								
5	H00346281	Scott	Gilbert	Ss2012@hw.ac.uk					x			x	
6	H00341198	Watt	Gregor	glw2001@hw.ac.uk					x				
7	H00344138	Legge	Stewart	Shl2000@hw.ac.uk				x				x	

Roles

Managing director	MD
Company secretary	Sec
Chief electronics and wiring engineer	EW
Chief digital systems engineer	DE
Software engineer	SE
Chief public relations organiser.	PR
Chief mechanical designer.	ME
Workshop engineer	WE
Web designer	WD

CHAPTER 2 BACKGROUND THEORY

In the modern world, where the requirement for more and more automation is higher than ever, one area where there is a significant demand is the healthcare sector. There are a wide variety of ways in which automation can be implemented, and one of these ways is the secure, automated transportation of medical supplies in hospitals. At first glance this may seem like a somewhat menial task to be focussing attention on, however once it is taken into consideration that where the already limited staff at hospitals and other medical facilities may need to dedicate critical time to moving supplies from location to location, if this system was automated staff could instead focus their attention on other extremely important tasks.

LITERATURE REVIEW: ULTRASONIC DISTANCE SENSOR THEORY

Ultrasonic distance sensors have many applications in robotics, by decreasing the need for human input it means they are being used even more prevalently in the healthcare industry. Ultrasonic distance sensors, as the name suggests, use ultrasonic waves to measure distance, by sending out these waves the reflection from them can be detected and therefore the distance of the object reflecting can be determined. The formula for this is:

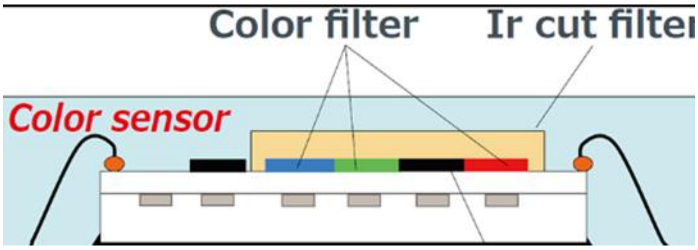
$$Dist. = \frac{1}{2} \times \ T \ \times \ C \ eq. 1 \quad [1]$$

Where T is the period between sending and receiving the waves and C is the speed of sound. There are many benefits to an ultrasonic distance sensor as they are incredibly well suited to many applications due to their ability to be used in all conditions. They can even work well in darker, dusty environments meaning their uses are greatly extended. Another benefit to them is their ability to detect clear surfaces such as water or glass/plastic, and even underwater as well (value of C must be changed for this to work to 4.3C).

Due to their low cost these sensors are also very accessible for people to use at home in their own projects or tests, making them a staple component to anyone learning programming on devices such as an Arduino. One downfall to these sensors is their ability to measure smaller more precise distances, however newer types have been developed for these sorts of uses but are not needed for most applications.

LITERATURE REVIEW AND BACKGROUND THEORY: COLOUR SENSOR

A typical colour sensor consists of 3-4 photodiodes covered by various filters. These diodes usually correspond to red, green, blue and an optional one for brightness. It is useful to have a white LED to illuminate the surface of the object the colour sensor is sensing as it means the photodiodes receive more consistent light levels in different environments. The diodes can differentiate between red, green and blue by having corresponding light filters on them which only let certain colours through. An infrared filter can also be used to block out unwanted IR that the photodiodes may pick up. Light filters on them which only let certain colours through. An infrared filter can also be used to block out unwanted IR that the photodiodes may pick up. [2]



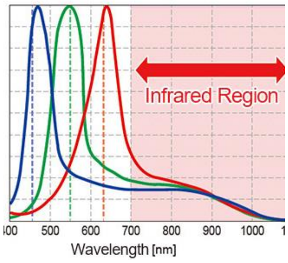


Figure 1 Detailed Illustration of Colour Sensor [2]

Figure 2 Wavelengths of Light [2]

An advanced colour sensor like the one used on the robot often has onboard microcontrollers to convert the values from the photodiodes into calibrated digital serial communications for a microcontroller. In our case the TCS34725 used I2C protocol.

A 2014 paper [3], goes into detail about characterising autonomous systems with colour sensors. The paper presents a system designed for an integrated circuit. The goal of the system was to reduce the complexity of a typical colour sensing system by using cheaper embedded components rather than combining multiple complex components.

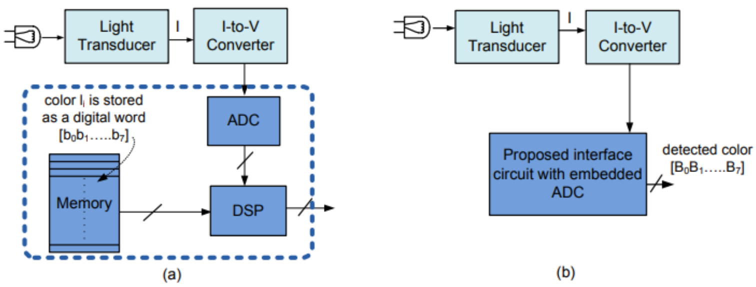


Figure 3 Block Diagram of Colour Detection Systems [3]

The proposed system was characterised on a breadboard and achieved better performance than typical systems under certain merits. This paper is relevant to the project as it highlights the project ethos of using cheaper and more reliable components for greater robustness.

LITERATURE REVIEW: PIEZOELECTRIC BUZZERS

Piezoelectric buzzers are simple devices used to generate sound tones. By applying an alternating electric signal to a piezoelectric material an audio signal can be produced [4]. When an electric current is applied to a piezoelectric material the shape of the material changes by expanding or contracting, this change in shape creates audio waves.

LITERATURE REVIEW: SERVO

Servos are small devices that are commonly used in robotics, automation, and control systems. They are essentially motors that are designed to rotate to a specific position and maintain that position.

A typical servo consists of three main components: a motor, a gearbox, and a control circuit [5]. The motor is usually a DC motor that is designed to rotate at high speed, while the gearbox is used to reduce the speed of the motor and increase its torque. The control circuit is used to regulate the position of the motor and ensure that it maintains a specific position.

When a signal is sent to a servo, the control circuit interprets the signal and determines the required position of the motor. The control circuit then sends a voltage to the motor that corresponds to the required position. The motor rotates until it reaches the desired position, and then the control circuit adjusts the voltage to maintain that position.

A website shows a labelled photo of a servo and explains its operation. The source also explains how sending different frequencies of signals to the servo changes the rotation of the 'horn'.

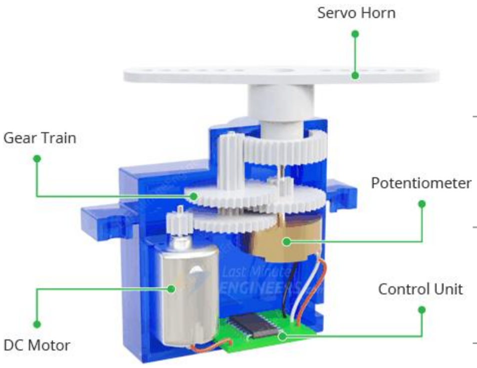


Figure 4 Construction of A Servo [6]

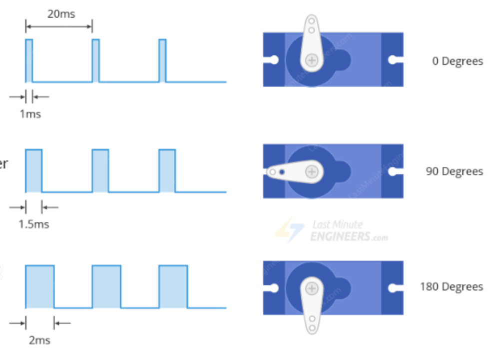


Figure 5 Pulse Width Controlling Servo [6]

This source is applicable to the assignment as we have used one to ensure that the medicine cabinet remains locked, as this was the simplest way we thought to do so.

MOTOR CONTROL AND PWM BACKGROUND THEORY

Motor control is the process of regulating the movement and speed of a motor. There is various control techniques to achieve precise control of a motor's position, velocity, and acceleration. In our robot, motor control is required to control the movement of the robot and we used Pulse Width Modulation (PWM) to do this.

PWM is a technique and form of motor control, it is used to control the speed of a motor by adjusting the average power delivered to the motor. It involves rapidly switching a signal on and off at a fixed frequency. Just like you can control the brightness of an LED by "flashing" it on and off at certain frequencies. The duration of the "on" time which is called the duty cycle, which determines the average power delivered to the motor. By changing the duty cycle of the signal, the speed of each motor can be controlled.

The robot's Arduino sends PWM signals to the motor driver, which then controls the amount of power delivered to the motors. By adjusting the duty cycle of the PWM signal, the speed and direction of the motors can be controlled, allowing the robot to drive forwards and turn, in this case in order to follow the lines of the hospital floor.

The reason we used PWM control is because of several advantages compared to other forms of motor control. Efficiency: PWM is a very efficient method of motor control because it only delivers the necessary amount of power to the motor to maintain the desired speed. This is important because the robot is running off of a relatively small battery. Precise Control: PWM allows for precise control of the motor's speed and direction, which is ideal for following the line closely and safely. Low Cost: PWM circuits are relatively simple and inexpensive to implement, which would allow us to keep costs down in production.

CHAPTER 3 METHODOLOGY/IMPLEMENTATION

To begin the design process there was firstly a meeting in which the members of the group discussed possible ideas for the bot. After looking over the specifications for the project and the resources provided, roles were allocated to the team members. It was at this point that the different members of the team began to work on conceptualizing the various aspects that would make up the bot.

CONCEPT DESIGN AND EVALUATION

Given the brief we chose to use the given platform instead of modifying it too heavily, we also used the components and tools that were readily available to create our solution.

This was not only easier due to availability, but all the components were also plentiful and cheap in case of errors. This also would make the robot easier to mass produce as no parts were hard to source.

DIGITAL AND SOFTWARE DESIGN

The Robot is Split in to a top and bottom half each controlled by an Arduino. Serial Communication is used to transfer the information about the colour that the Robot should detect from the top half Arduino to the bottom half Arduino.

The Lower Half Algorithmic state machine Diagram Shown in the figure below describes the operation of the Robot as controlled by the lower half Arduino. The Arduino attempts to determine the location of the line and turns left, right or straight ahead depending on its location, as determined by the sensors. If an object is detected as obstructing the Robot’s path the robot stops moving and waits till the obstruction has been cleared. The Arduino then detects the colour of the surface it is on and turns left if the colour is the same as the selected colour which was communicated to it by the top half through serial communication. If the colour of the surface is Green, then the robot has reached its destination and stops moving.

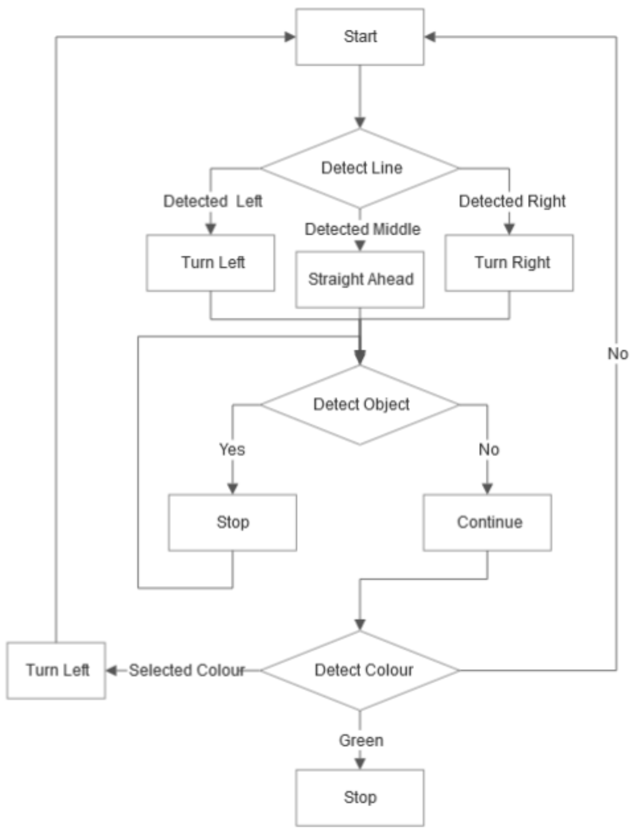


Figure 6 Lower Half Algorithmic State Machine

The Algorithmic state machine diagram of the Top half Arduino is shown in the figure below in two parts. The Diagram shows the selection of a colour, the setting of a pin code. After the pin code has been set the cabinet locks if a package is in the cabinet. The pin code is then entered again to unlock the cabinet if the code is correct.

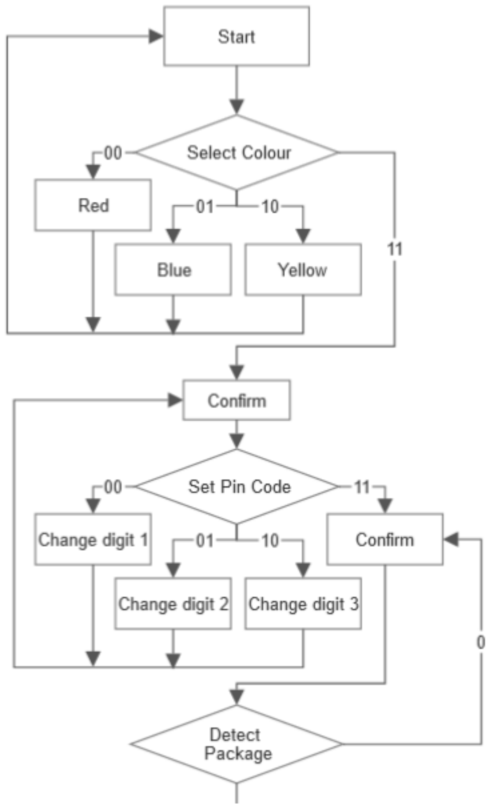


Figure 7 Top Half Algorithmic State Machine Part 1

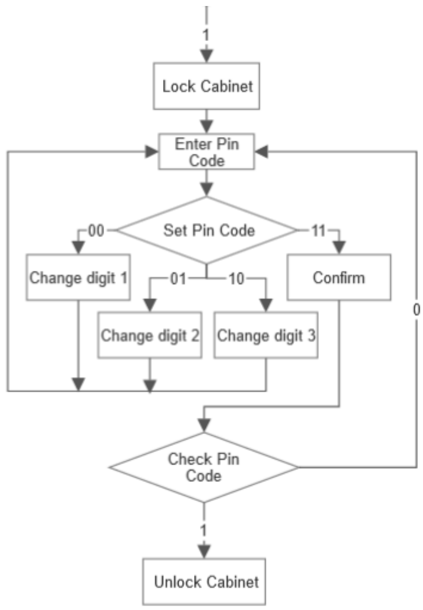


Figure 8 Top Half Algorithmic State Machine Part 2

LIST OF FUNCTIONS WRITTEN

Functions from code taken from the course notes have not been included in this list.

Top Arduino Functions

- readButtonState()
- lock()
- unlock()
- selectColour()
- digitCodeSelection()
- checkDigitCode(int setCode[3], int inputCode[3])
- checkPackageInCase()
- lockPackageInCase()

Bottom Arduino Functions

- colorDistance(int r, int g, int b, int color[])
- detectColour()

Some components such as the LCD screen and the colour sensor use the I2C communication protocol to transfer data back and forth between the Arduino boards and components.

The SoftwareSerial library was also used to allow communication between two Arduino boards. By defining pins to be used for communication the Arduino boards were able to transfer data such as the top Arduino board communicating the ColourID of the colour that is the destination for the robot, to the bottom board.

In testing the two Arduino board were able to communicate using the SoftwareSerial library, however when attempting to implement this alongside the rest of the functionality of the robot the communication failed Due to being unable to fix this issue a button was added to the bottom half of the robot to cycle through the possible colours.

USER INTERFACE

The Robot displays the user interface on an LCD panel placed at the front of the robot. The user interface initially displays the chosen colour of the destination, each of the possible colours are shown below.

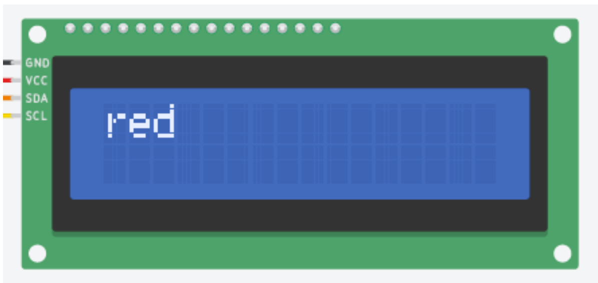


Figure 9 LCD Screen Displaying text “red”

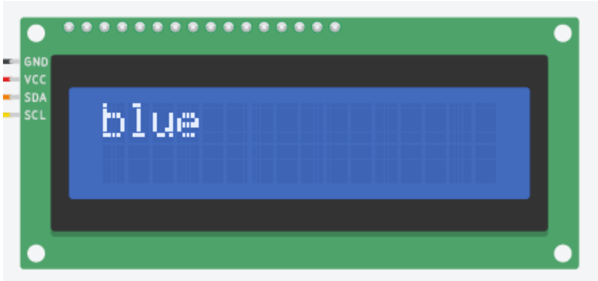


Figure 10 LCD Screen Displaying text "blue"

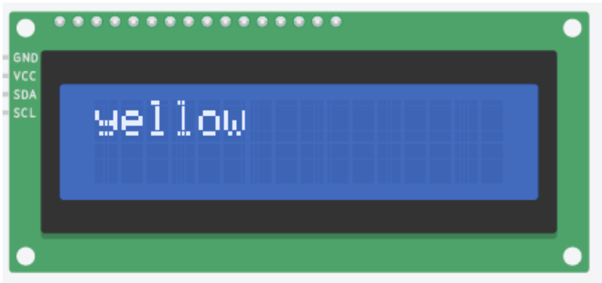


Figure 11 LCD Screen Displaying text "yellow"

After a destination has been selected by its colour the LCD screen displays the 3-digit code that is to be set, with the initial value of this code being 0 0 0.

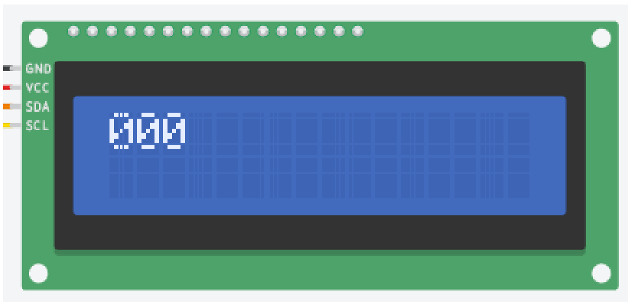


Figure 12 LCD Screen Displaying 3-Digit Lock Code

When the code is set the screen displays a message asking for the package to be placed in the compartment and the lid closed, after this a closing message is displayed during the 10 second delay between a package being placed and the lid locking.

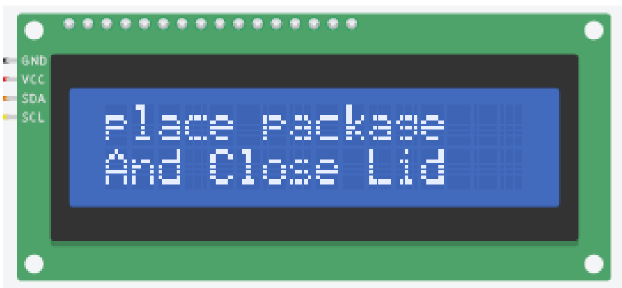


Figure 13 LCD Screen Displaying Place Package Message

Figure 14 LCD Screen Displaying Closing Message

After the lid is locked the LCD screen displays the code used to unlock the cabinet, and then it displays an unlocked message.

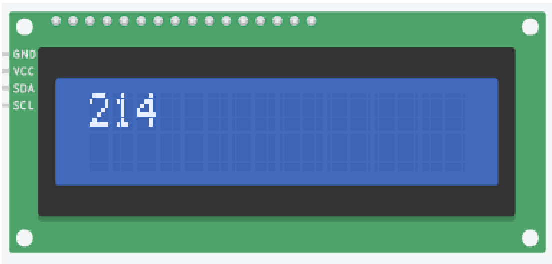


Figure 15 LCD Screen Displaying 3-Digit Unlock Code

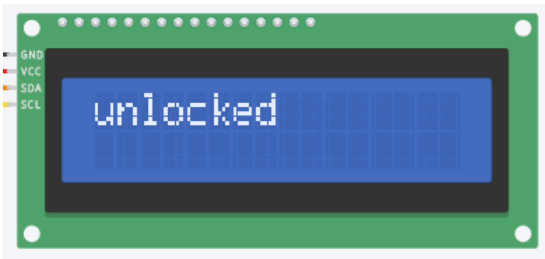


Figure 16 LCD Screen Displaying Unlocked Message

COLOUR SENSOR CALIBRATION SOFTWARE DESIGN

The colour sensor was required to detect non-trivial colours outside of just red, green and blue, it also needed to correctly detect a non-uniform floor pattern, so it was necessary to create software that was able to differentiate between colours. This was done by treating each rgb value received as a point in space and by also having a set of points that represented the require colours in that space. The distance was taken between these the sensor point and each colour point and the lowest distance represented the color detected.

$$\text{distance} = \sqrt{(R_2 - R_1)^2 + (G_2 - G_1)^2 + (B_2 - B_1)^2}. \quad \text{[eq.2]}$$

SENSOR SYSTEM BLOCK DIAGRAM

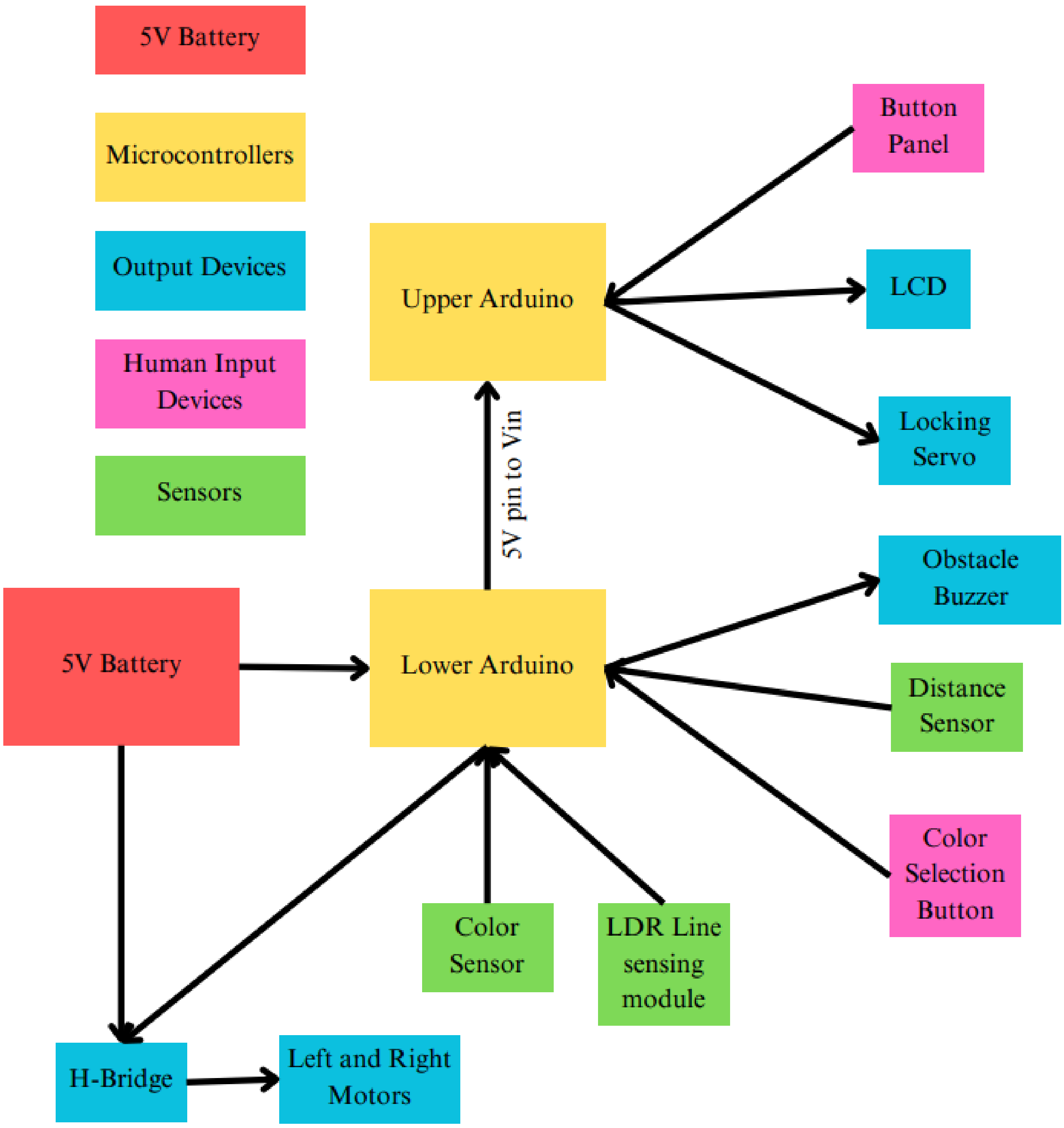


Figure 17 Block Diagram of Sensor System

CIRCUIT SCHEMATIC

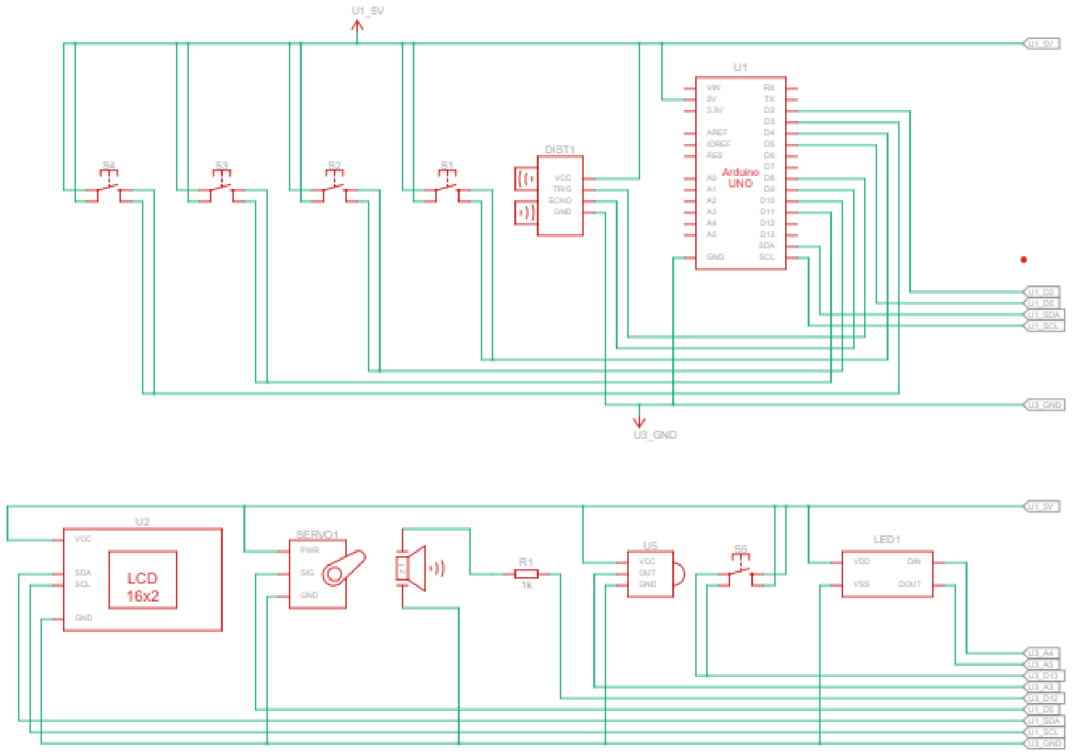


Figure 18 Circuit Schematic Top Half

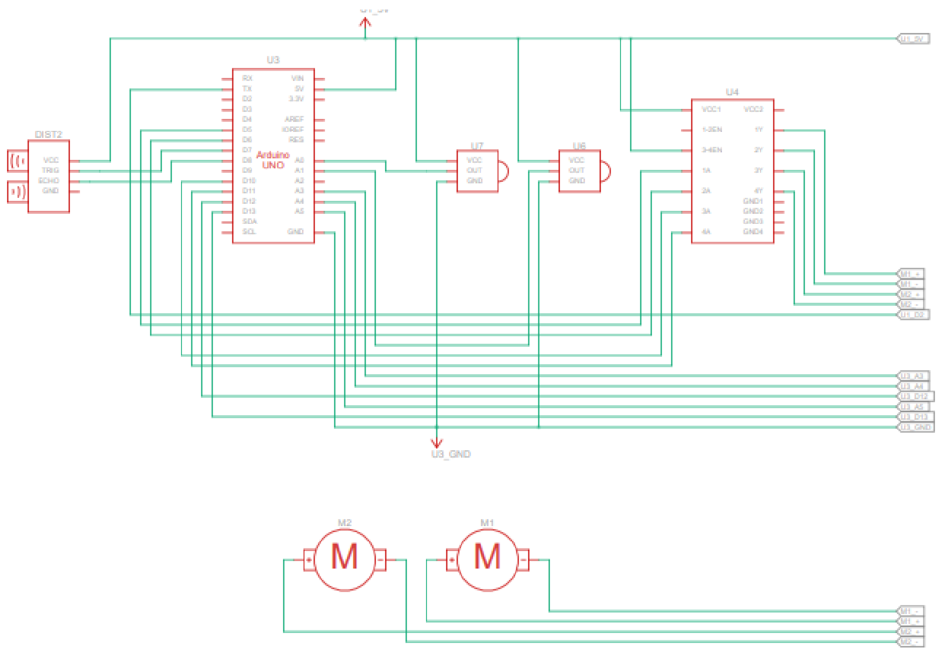


Figure 19 Circuit Schematic Lower Half

MECHANICAL DESIGN

The main mechanical components of the robot are the locking mechanism and the hinged lid for the compartment for the medication. The locking mechanism comprises of these components:

- Servo
- Metal L-bracket
- Screw/Nuts
- The lid with cut-out

Figure 20 Top View of Hinged Lid

Above is a rough preliminary sketch of the lid, which was slightly adapted for the final design. The mechanism is designed to work by the L-bracket being moved within the L-shaped slot in the lid. At full upright and at 90º perpendicular to the lid, the compartment would be locked as the L bracket will stop the lid from opening. Once the servo is turned approximately 10º (so the L-bracket is at an angle) the lid can be lifted due to the lids cut-out having a larger gap

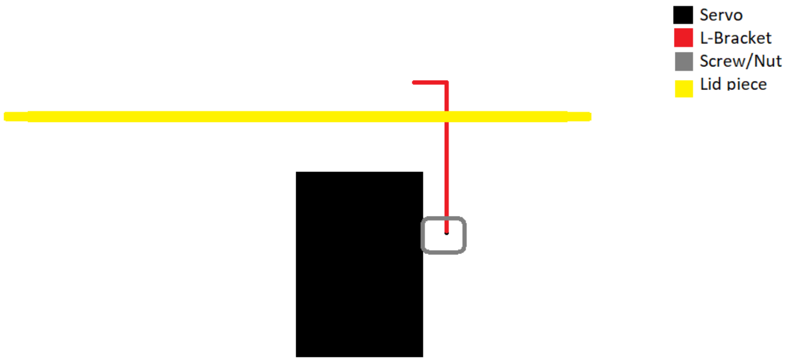


Figure 21 Side View of Hinged Lid

This is a sketched view of the mechanism from the side and shows how the pieces all slot together, with the servo sitting underneath inside the main compartment of the robot, and the L-bracket attached to it via the screw.

MODELLING AND PROTOTYPE

When designing our overall goal was to keep the robot as simple and compact as possible, while also achieving everything set out by the brief. This means our design uses the pre-given base for the robot and is built around it, with a series of layered shelves to house the Arduinos and breadboards, and a large split compartment that is placed on top, where 3/4s of the compartment is for medication and the remaining 1/4 is for housing the servo for the lock, as well as a ultrasonic distance sensor, the back of an LCD and all the wiring for them.

Our first sketch was again kept relatively simple, and we built upon it to create the final product, here it is shown below:

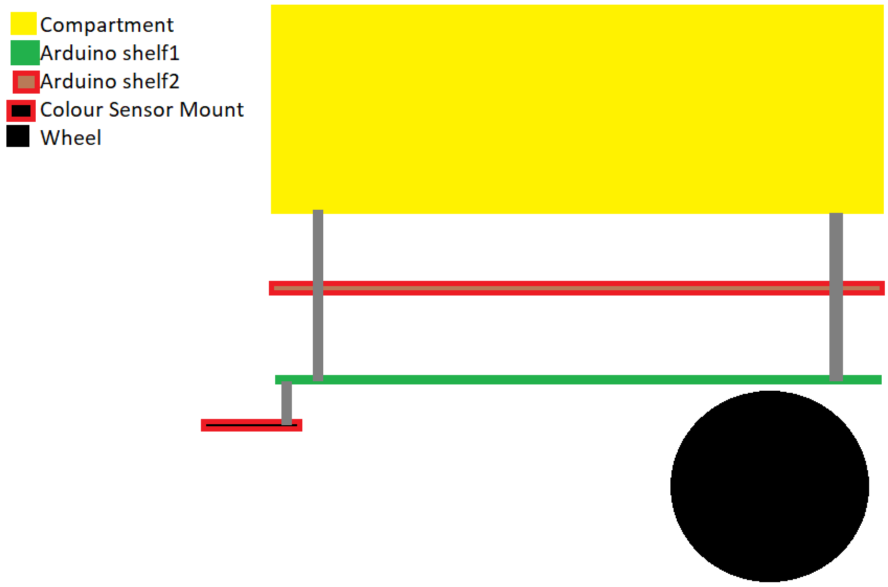
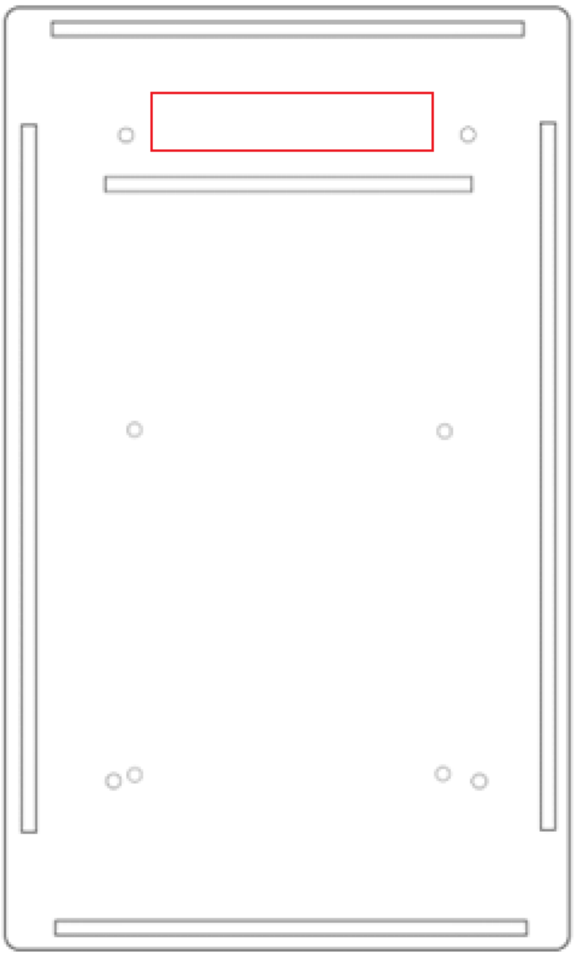


Figure 22 Sketch of Prototype Design

Our first goal from this was to then design the pieces needed for this to be possible. This was done on 2 different software, Inkscape and Inventor.

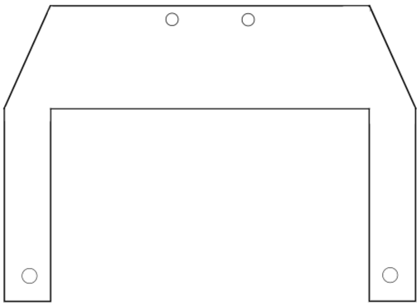
The shelf/bottom of compartment:



This piece was designed to be able to be used as the base of compartment and the Arduino shelf. The large slots along the edges and middle of the piece are dual purpose with them being used to pass wires up through for the Arduino shelf and they are also used for the walls of the compartment to slot into meaning the compartment is fully interlocking. The only difference between the 2 is that the compartment base will have the red box also cut-out while the shelf doesn't, this is so wires can be passed up to the servo, distance sensor and LCD.

Figure 23 Shelf and Bottom Cut-out

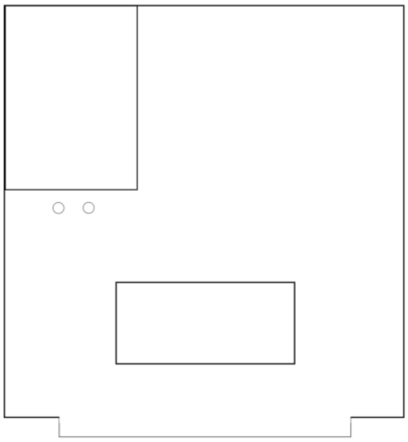
Colour sensor mount:



This mount attaches to the front of the robot using spacers hanging from the Arduino shelf1, it will straddle the line following attachment at the front and the colour sensor will be screwed on using the top 2 screw holes of the mount.

Figure 24 Sensor Mount Cut-out

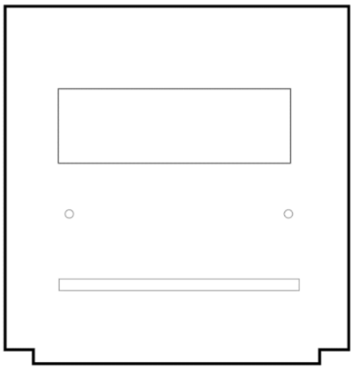
Inner wall of compartment:



This inner wall will slot inside the compartment separating the components and their wiring from the medication. The cut-out on the top left will be where the servo sits with the two screws holes underneath holding it in place, the second cut-out is for the distance sensor to sit in, which will detect if there is any medication in the compartment.

Figure 25 Inner Wall Cut-out

Front piece:



This front wall of the compartment will have the LCD display in the larger of the cut-outs and the smaller will hold the buttons for the entering the PIN for the robot and controlling it in general.

Figure 26 Front Panel Cut-out

Back Piece:

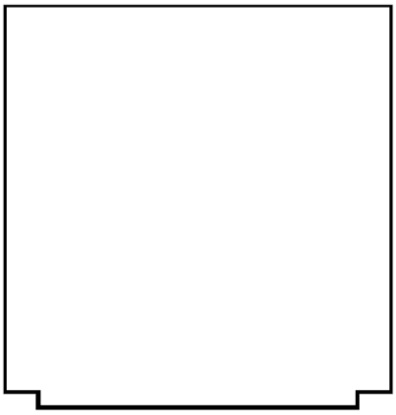


Figure 27 Back Panel Cut-out

Side Pieces:

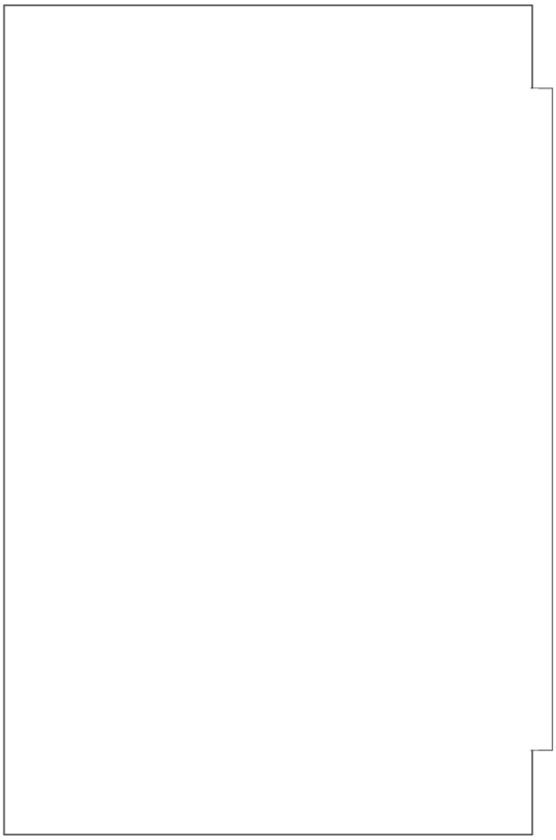


Figure 28 Side Panel Cut-out

When all slotted together they should create something along the lines of this:

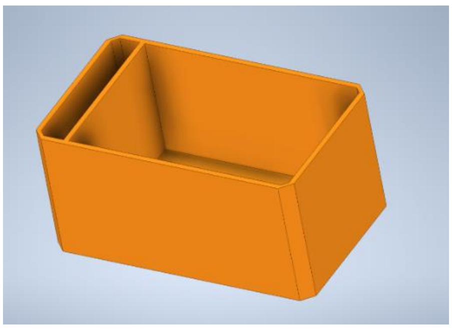


Figure 29 Compartment Prototype CAD Model

A lid is also needed and when added looks like this:

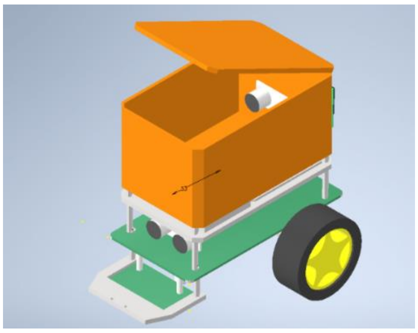


Figure 30 Robot Prototype CAD Model

Constructing the Robot

A few changes were made between the CAD model and the actual robot, for example the LCD was placed on the front rather than the back (as well as the locking mechanism) and the spacing between each shelf was increased to fit the components together more easily, which resulted in a slightly taller design.

The main way of creating the pieces was to use 3mm Acrylic on the laser cutter, which takes the .svg files created in Inkscape and cuts them out. We mainly tried to use scrap acrylic as well, hence the multicoloured design as we believed it would be the best thing for us to do to make the project as least wasteful as possible.

The extra height had little to no effect on the robot's stability or performance and was a needed addition for the robot's functionality. The spacers use to separate the shelves and the compartment where a mix of 3mm metal and nylon spacers (nylon needed when touching PCB boards to avoid damage). The compartment was held together using strong plastic glue on top of the interlocking pieces ensuring it was as structurally sound as possible.

Figure 31 Picture of Assembled Robot

This picture shows the completed Robot.

Figure 32 Picture of Compartment Internals

This picture shows the inside compartment of the robot, where you can see how it is split between the main compartment and the with all the components and locking mechanism.

CHAPTER 4 RESULTS AND DISCUSSION

COMPUTER SIMULATION

This Tinkercad simulation was used to experiment the different features our robot car would have. The circuit shown in Figure 1 represents our robot car with various components, including Arduino boards, sensors, motors, an H-bridge, an LCD, LEDs, resistors, and pushbuttons.

The main Arduino board is connected to an H-bridge module, which powers and controls the two DC motors that drive the car's movement.

The car is also equipped with several sensors that provide information about the environment. There is an ultrasonic sensor that measures distances for obstacles detection, a colour sensor that senses different colours, three IR sensors that detects the line it is supposed to follow, and a servo motor that would be used as a lock for the medicine compartment. The LCD displays information about the status of the robot's compartment and the paths the user chose. Pushbuttons are used for the pin that unlocks the box and to toggle between the different displayed paths.

The circuit also includes several LEDs and resistors, which wouldn't be used on the physical robot, but provide feedback and help us control the car's behaviour.

The code for this circuit is found using the Tinkercad link shown below in figure 2. This code will instruct the robot to first scan for black line using the IR sensors and updates the direction of the motors based on the readings of the sensors by setting the speed of the DC motors. It then uses the ultrasonic sensor to detect if there are obstacles in front and if there is an obstacle detected within 12cm, the motors will stop. The values of the sensors are also printed on the serial monitor for analysis purposes.

The code and circuits for robot navigation and obstacle avoidance can be tested and improved with the help of this simulation. It enables the testing of the circuit and the code in a secure and controlled environment before applying it to a real robot.

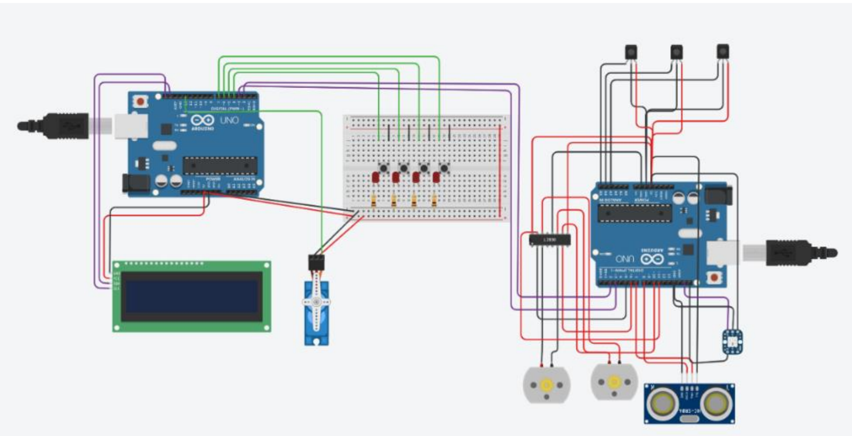


Figure 33 TinkerCAD Model of Robot

RESULTS AND DISCUSSION

Results

Throughout the semester we built a robot car capable of following lines, avoiding obstacles, detecting colours, and locking a compartment depending on the pin input. The robot successfully performed all these tasks during our presentation: the robot managed to follow a black line set on a white table and turn when it detected a colour (red, blue or yellow), that represented the possible paths, chosen by the user and then stopping when it reached a green square which represented the destination. And only then was the pin input and the compartment unlocked. After being manually flipped, the robot followed the same line and turned left when it detected the same colour and just stopped when it got to another green square, this time, representing the base station. During this presentation we also tested the obstacle detection system by moving things in front of the robot causing it to stop and sounding the buzzer, as expected.

During testing we found that the line following and sometimes the colour sensing mechanisms were not well calibrated, and the lighting in the room wasn't ideal for sensing, but the robot's overall performance was more than satisfying and it was found to be consistent, reliable, and efficient, making it suitable for use in hospital settings.

Discussion

In general, we were pleased with the results of our project. The robot was able to perform all the tasks we had designed it for, although there were some minor issues related to calibration, as mentioned above. Proper calibration of the sensors would be critical to achieve accurate and reliable results. We believe that with further refinement and testing, the line following and colour sensing mechanisms of the robot could be made more accurate.

Our project successfully demonstrated the feasibility of building an Arduino-based robot car capable of line following, obstacle avoidance, colour detection, and compartment lock unlocking. We identified areas for improvement, particularly in sensor calibration and powering.

During the early stages of the testing phase, another issue related to the colour sensor arose: it wasn't detecting colours, or it was detecting the wrong colours, when it was being power by the power supply unit. But when connected to a computer, it would work perfectly. We modified the mechanism several times but could not solve the problem, until we realised that it was just a wire that wasn't well soldered. After fixing that issue the sensor worked well again. Another limitation of our project was the time constraint. We were only given a limited amount of time to complete the project, which meant that we had to prioritize certain aspects over others. If we had more time, we would have liked to fine-tune all the components of the robot to make it more accurate and efficient. We also attempted to use limit switches in the compartment to detect if there was any medicine inside it, but they were too sensitive. We ended up using an ultrasonic sensor, which worked very well.

In conclusion, we believe that our robot car was a success, despite the minor issues related to calibration. We were able to demonstrate that the robot could perform the tasks we had designed it for, and we believe that with further refinement and testing, it could be a useful tool in a hospital setting or other similar environments.

CHAPTER 5 MANAGEMENT AND MARKETING

For our company name we decided G-43 (the name of the bench we were working at in the lab), we believe this is a professional name that works well due to its simplicity. The name for our robot was decided to be ESCOBOT, a quirky inventive name that we hope will stick in the mind of anybody that sees it.

FLYER

Figure 34 Front and Back Sides of Flyer

This first side is the flyer design for the robot (front), with details of our presentation. The other side gives a brief explanation of the robot and its uses.

We decided on this colour scheme as we believed the darker blue background really brought out the bright yellow of the text and robot, making them pop out the page. This can be confirmed using colour theory where blue and yellow (plus the slight red from the robot), are extremely complimentary colours with the blue allowing the yellow to stand out and grab the attention of the reader.

WEBSITE

Figure 35 Website Front Page

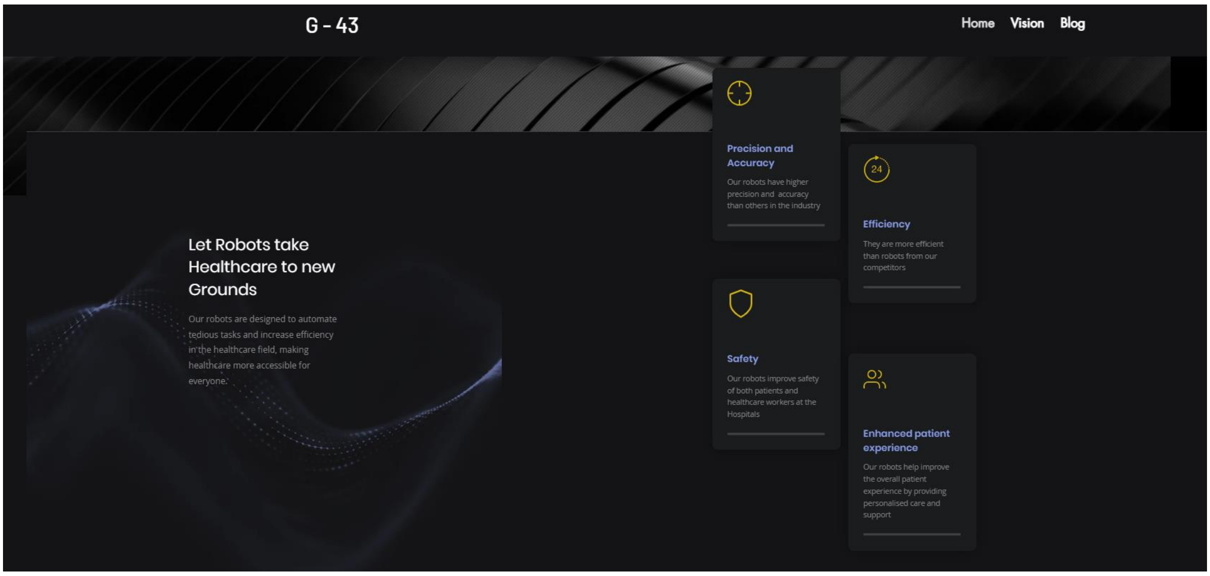


Figure 36 Home Page Details

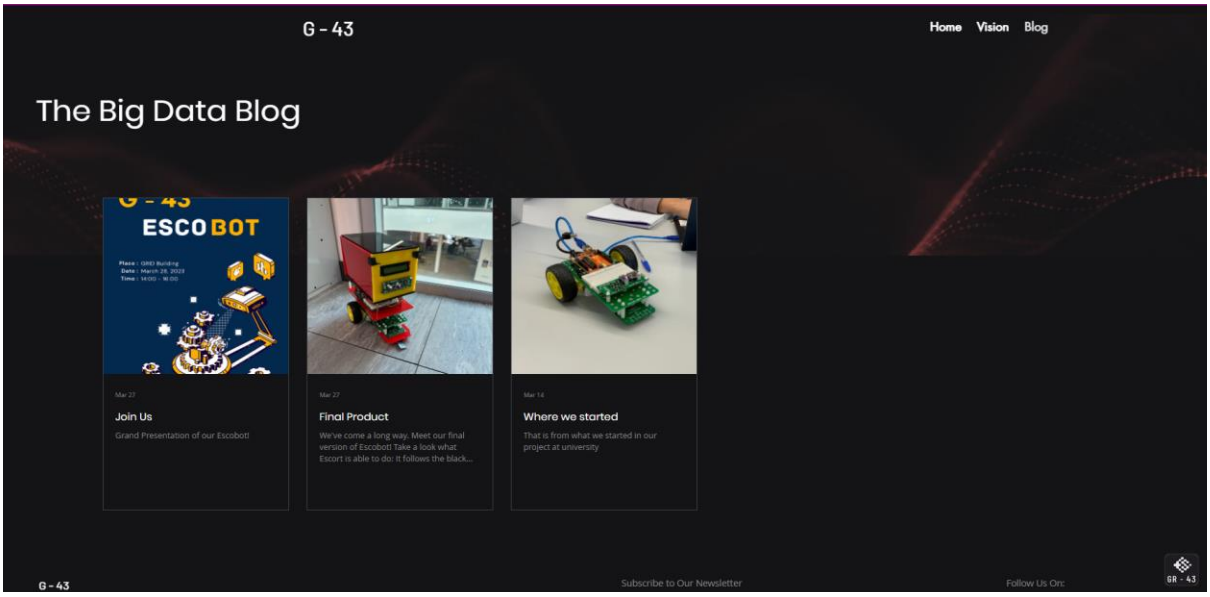


Figure 37 Website Blog Page

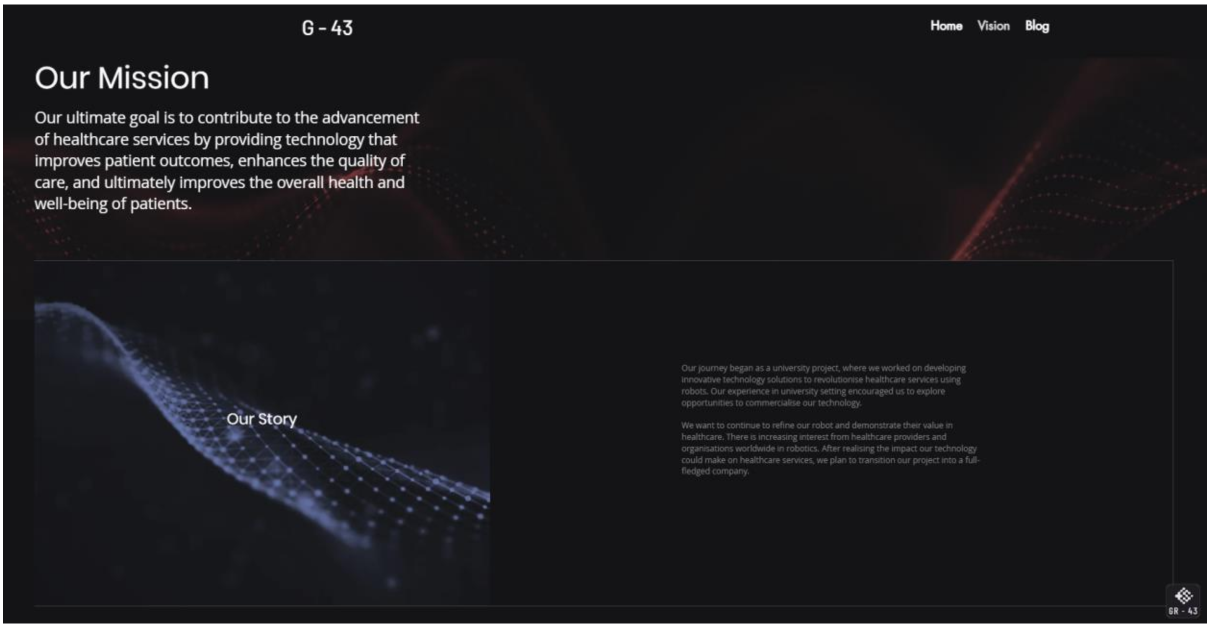


Figure 38 Our Mission Website Page

Our theme for the website was darker colour theme with accents of the three primary colours in colour theory (red, blue, and yellow). We believe this creates a slick professional feel to the website, something that could be trusted and relied upon in the field of medicine.

We decided to also include a blog to the website so anyone interested in the product can see its progress and any new features or design that we implement. We also had a page showing our vision and the goals the company has set out from the start and how we believe our product can benefit the field of medicine.

SWOT ANALYSIS

Figure 39 SWOT Analysis

RISK MANAGEMENT

We took many steps in order to minimise risks in the construction and operation of the robot.

During the construction we ensured that any of the workshop engineers were trained to operate any of the machines that were used to create the components of the robot, for example, the Laser Cutter and ensured that vices and safety goggles were worn when required.

For the operation of the robot, we included several features to minimise risks to anybody near the robot. First of all, we made sure that the robot always moved fairly sedately, we also included a buzzer to notify people of the robot as it is low to the ground and finally, we included ultrasonic distance sensors that would notify the robot to slow down and stop if a person was in front of it to minimise the risk of collision. A risk assessment for the laser cutter was carried out and is in the appendix, highlighting all the major issues that could have arisen when in use.

CHAPTER 6 CONCLUSIONS AND FUTURE WORKS

CONCLUSIONS

We were successful in meeting the brief in almost every aspect. We created a proof-of-concept robot which safely and reliably delivered the medicines to the correct place. We successfully simulated said robot beforehand and the robot met all the required specifications. We worked well as a team to do this pooling our collective skills and strengths.

FUTURE WORKS

In the Future there are some small issues that could be resolved within the software and schematics of our robot. Fixing the communication between the two Arduino boards and implementing more advanced inputs and outputs for the benefit of the end user experience. We would also be investigating a more secure locking mechanism to ensure that the cargo is left safe and untampered in transit.

REFERENCES

Referenced in IEEE style

[1] Keyence "What is an ultrasonic distance sensor" Keyenceco.uk. https://www.keyence.co.uk/ss/products/sensor/sensorbasics/ultrasonic/info/#:~:text=As%20the%20name%20indicates%2C%20ultrasonic,between%20the%20emission%20and%20reception. (accessed Apr, 26, 2023)

[2] ROHM Semiconductor. "Color Sensor" Rohm.com. https://www.rohm.com/electronics-basics/sensor/color-sensor (accessed Apr. 26, 2023).

[3] Maher Assaad, Israel Yohannes, Amine Bermak, Dominique Ginhac, Fabrice Meriaudeau. "Design and Characterization of Automated Color Sensors System." https://www.researchgate.net/publication/281659985_Design_and_Characterization_of_Automated_Color_Sensors_System (accessed Apr. 27, 2023).

[4] APC International, Ltd. "Piezo Buzzers" Americanpiezo.com. https://www.americanpiezo.com/standard-products/buzzers.html (accessed Apr, 27, 2023).

[5] KOLLMORGEN "How Servo Motors Work" KOLLMORGEN.com. https://www.kollmorgen.com/en-us/blogs/how-servo-motors-work (accessed Apr, 27, 2023).

[6] Last Minute Engineers "How Servo Motor Works & Interface It With Arduino" LastMinuteEngineers.com. https://lastminuteengineers.com/servo-motor-arduino-tutorial/?utm_content=cmp-true (accessed Apr, 27, 2023).

APPENDIX

APPENDIX A RISK ASSESSMENT

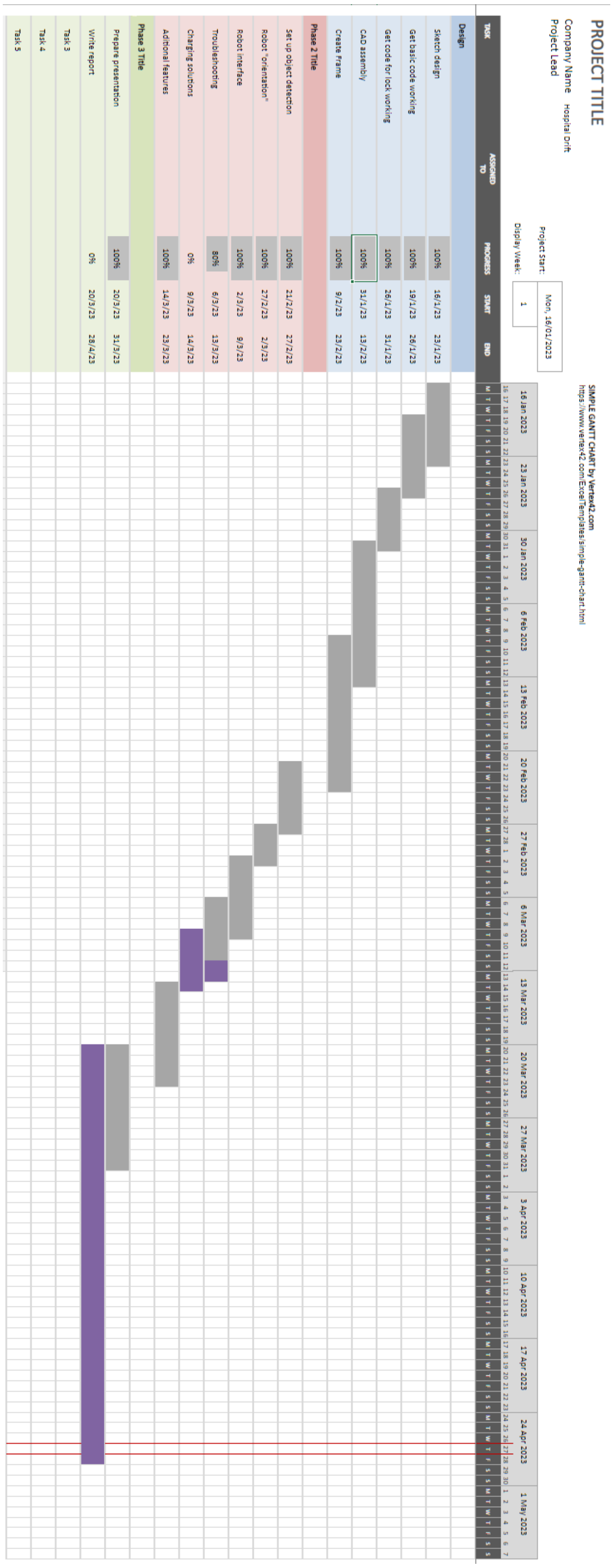
Chemicals		Oils/Lubricating Materials		Complete a COSHH form for chemicals.
Solvents		Gases or Liquid Gases		
Biological Substances		Powders or Dust	X	
Glues/Adhesives		Fume emissions	X	
Radio-active Sources		X-rays		
UV		Microwave or RF Sources		
Visible or Non-Visible Lasers			X	
Asbestos				
All Electrical Equipment			X	Ensure Equipment is within its test period

-AC or DC Supplies over 50 Volts	X	-Use of Display Screen Equipment (2 hours per day or more most days)	
-Servicing Electrical Equip. or Supplies	X	-Gas under Pressure	
-Hot surfaces or Ovens		-Pressure/Vacuum Systems	
-Flammability, Fires	X	-Cryogens	
-Welding		-Manual Handling	
-Motors & Rotating Machines		-Use of Slings or Shackles	
-Power Drilling		-Use of Lifting Equipment or Jacks	
-Lathe Work or Milling		-Working at Height	
-Sharp Edges or Implements		-Water tanks	
-Noise		-Diving	
		-Others – Specify below	

The One-Page Risk Assessment - Laser Cutter

Identified Hazard	**Specific Control Measures** e.g. Safety Interlock, Personal Protection Equipment or Clothing (goggles, face-mask, safety foot-ware, etc). Are written procedures are available?	**Residual Risk** Low, Med, High
Laser, could cause fire to acrylic, melted acrylic could then burn through clothes	Wear lab coat and always keep an eye on the laser cutter so a fire can be extinguished immediately	LOW
Dust/scraps could light on fire	The machine should be brushed down after every use to ensure it is clean and free of scrap pieces that could light.	LOW
Fumes from burnt materials	Turn extractor fan on before every use	LOW

APPENDIX B GANNT CHART



Meeting Minutes			
Group Name:	Hospital Drift	**Team Number:**	06

Week:	Week 1	
Time:	1400	
Place:	GRID	
Present:	Joao Lopes Gilbert Scott Iain Harkness Kieran Baird Alexandra Cures	
Apologies:		
Agenda:	• Meet the team. • Collect project kit. • Discussed team name. • Allocated team roles. • Discussed project specifications.	
Review of previous actions:	-	
Main business:		**Action**
First team meeting – allocating team roles and	Team roles were allocated as follows: Joao – Managing director and chief digital systems engineer. Kieran – Company secretary and workshop engineer.	

discussing project:	Iain – Chief electronics and wiring engineer. Alex -Chief public relations organiser and web designer. Gilbert – Software engineer.
Discussed design specifications :	The project specifications for the project were looked over, as each design requirement was discussed in detail.
Discussed possible meeting times:	After a discussion about what times the team members were available on their timetable, it was decided that the best time for additional meetings out with timetabled sessions would be on Mondays at 11am.
In preparation for next meeting:	It was agreed that all group members would do general research about the project in preparation for the next meeting.

Meeting Minutes			
Group Name:	Group 6	**Team Number:**	6
Week:	Week number 3		
Time:	When 30/01		
Place:	Where GRID		
Present:	Who was present at the meeting Iain Alex		

	Gregor Stewart Joao	
Apologies:	Gilbert Kieran - Work	
Agenda:	Get Line following code working If line following works, try out code for distance sensor	
Review of previous actions:	Specific reporting of actions from previous meeting. Person identified as responsible for the action to provide the report	
Main business:		**Action**
Line Follow & Distance sensor	With slight tweaking of the code the robot was finally able to follow a lined path. This was possible by switching the how the motors reacted to the LDR's. We found that slowing down the speed of the motors helped as well as moving the robot to a flatter smoother surface, as the wheels of the robot were struggling when going over bumps that were on the mats provided. We also attached the distance sensor, however this made the line following code not work at first, as it slowed the detection speed from the LDR's.	

Meeting Minutes			
Group Name:	Pablo Escobot	**Team Number:**	06
Week:	Week 3		
Time:	1300		
Place:	GRID		
Present:	Joao Lopes Gilbert Scott Iain Harkness Kieran Baird Alexandra Cures Gregor Watt		
Apologies:	Stewart Legge		
Agenda:	• Work on code for bot • Find dimensions of bot and develop CAD model • Discuss actions for next meeting		
Review of previous actions:	Work had been done on the bots code, and methods for wiring the bot were discussed A Cad model was created based on the dimensions of the bot taken during the last meeting		
Main business:		**Action**	

Work was done on the bot code:	There was discussion about the optimal way the bot could be configured, and it was discussed that an LCD could be used in the bots design.
CAD model was worked on :	Further measurements were taken on the exact dimensions of the bot to further detail the CAD model
In preparation for next meeting:	-Looking into obtaining an LCD for the bot, and possible ways of using an LCD -The incomplete CAD model will be worked on further, adding detail to base design

Meeting Minutes			
Group Name:	6	**Team Number:**	6
Week:	Week number 5		
Time:	When 14/02/2023		
Place:	Where GRID		
Present:	Iain Joao Stewart Gilbert		
Apologies :			

Agenda:	Start assembly of Robot structure so second Arduino can be placed and used in the design Figure where wheel could be placed at the front of the robot	
Review of previous actions:	(Iain) CAD – I finished the initial CAD design, so we could start the construction of our Robot, I do not expect the design to be final but more just a rough guide for us to follow when starting our construction. It will comprise of two shelves for the arduinos, then a further third for the compartment on top, which will store the drugs/medicine.	
Main business:		**Action**
	Began the assembly of the robot and secured the design for the shelf for the second Arduino. We also tested out how we could implement a freely rotating front wheel, but we are unsure if we will proceed with it as it looks like it may lift the LDR's to far off the ground affecting the robots line reading capabilities.	

Meeting Minutes			
Group Name:	Group 6	**Team Number:**	6
Week:	Week number 7		
Time:	When 28/02		
Place:	Where GRID		
Present:	Who was present at the meeting:		

	Joao Iain Gregor Gilbert Stewart Kieran
Apologies:	
Agenda:	Finishing CAD assembly
Review of previous actions:	We did not meet in week 6 but everyone worked on their CAD parts at home as planned.

Main business:		**Action**
	The present group members finished off their CAD parts. Iain did the Robot Base/Wheels/compartment, Gilbert the Colour sensor Mount/LCD, Gregor the top shelf, Stewart the Nuts/Bolts and Joao the Ultrasonic Sensor/lid. Stewart assembled all the parts for the final design. Iain submitted the assignment. <u>For next meeting:</u> Gilbert will finish the code for the locking mechanism. Start planning and designing the box pieces that need to be laser cut and assembled. Iain will get training on the laser cutter.	

Meeting Minutes

<table>
<tr><td>Group Name:</td><td>Group 6</td><td>Team Number:</td><td>6</td></tr>
<tr><td>Week:</td><td colspan="3">9</td></tr>
<tr><td>Time:</td><td colspan="3">1400</td></tr>
<tr><td>Place:</td><td colspan="3">GRID</td></tr>
<tr><td>Present:</td><td colspan="3">Kieran Baird
Iain Harkness
Alexandra Cures</td></tr>
<tr><td>Apologies:</td><td colspan="3">Gilbert Scott,</td></tr>
<tr><td>Agenda:</td><td colspan="3">Robot Demonstration
Meeting with project supervisor
Final report draft</td></tr>
<tr><td>Review of previous actions:</td><td colspan="3">Iain has started on a Wix website for the PR section of the project.
Kieran has began to write a draft for the final report.</td></tr>
<tr><td>Main business:</td><td colspan="2"></td><td>Action</td></tr>
<tr><td></td><td colspan="3">The progress made on the Wix website was continued by Iain and Alexandra, and this work will continue for future meetings. The password for the wix account was shared with the rest of the team.
The deadlines were discussed by Alexandra and Kieran, and how work can progress forward to meet upcoming deadlines.</td></tr>
</table>

	The team met with the project supervisor, who gave advice on upcoming deadlines, better meeting minutes and organisation, and more effective ways to progress.

Meeting Minutes			
Group Name:	Group 6	**Team Number:**	6
Week:	9		
Time:	1400 9/3/23		
Place:	GRID		
Present:	Kieran Baird Gilbert Scott		
Apologies:	Stewart Legge Alexandra Cures		
Agenda:	Progress on the website and final report draft Recap meeting with project supervisor Discussing progress on the bot		
Review of previous actions:	The final report draft has been typed up and put on MS Teams The bot now has a second platform that has been made for an additional Arduino and breadboard, and this was subsequently wired.		
Main business:		**Action**	

	Points for previous meeting was recapped for Gilbert, who could not attend. The progress on the bot was discussed, and it was decided that there needed to be additional parts made for the front of the bot in order to keep some components in place. It was discussed whether it would be more efficient if there was several smaller parts fabricated and glued together rather than one large piece being printed. The website was discussed, with ways in which to improve it. Work will continue on writing the code that will allow the two different Arduinos on the bot to communicate with one another.

Meeting Minutes			
Group Name:	Group 6	**Team Number:**	6
Week:	Week number 9		
Time:	When 14/03		
Place:	Where GRID		
Present:	Who was present at the meeting: Joao Iain Alex Gilbert Stewart Kieran		
Apologies:	Gregor		
Agenda:	Discuss laser cutting components Decide how to proceed with upper platform		

<table>
<tr><td></td><td colspan="2">Detection of obstacles by the bot code</td></tr>
<tr><td>Review of previous actions:</td><td colspan="2">It has been decided that it would be best if the required components to mount the sensor were cut in smaller parts and glued together.

A schematic has been drawn up for the sensor at the front of the bot.

The flyer has been completed

Gilbert has completed and simulated code for the locking system</td></tr>
<tr><td>Main business:</td><td></td><td>Action</td></tr>
<tr><td></td><td colspan="2">Iain checked the availability for laser cutters to cut out the components required to mount the sensor, there is no time available for today, Tuesday 14/3, so a time on Thursday 16/3 will be booked instead.

Gilbert and Iain are designing components for the top shelf for the cargo to be cut on Thursday along with the already drawn up plans for the front sensor.

Iain has booked the laser printer for Thursday.

Alex has made significant progress on the website, various pictures will be uploaded at different stages in development for the ‘blog section’ part of the website.

<u>For next meeting:</u>

Iain, Gilbert and Stewart will complete the design for the shelf to be cut out by the laser on Thursday.

Alex plans to finish the website this week and start working on the finance report.</td></tr>
</table>

<table>
<tr><td>Group Name:</td><td>Group 6</td><td>Team Number:</td><td>6</td></tr>
<tr><td>Week:</td><td colspan="3">Week number 10</td></tr>
<tr><td>Time:</td><td colspan="3">When 21/03</td></tr>
<tr><td>Place:</td><td colspan="3">Where GRID</td></tr>
<tr><td>Present:</td><td colspan="3">Who was present at the meeting:
Joao
Iain
Gregor
Gilbert
Stewart</td></tr>
<tr><td>Apologies:</td><td colspan="3">Kieran</td></tr>
<tr><td>Agenda:</td><td colspan="3">Laser cutting components for the bot..
Continue work on the code and the hardware for the bot.
Fix power issue</td></tr>
<tr><td>Review of previous actions:</td><td colspan="3">The issue with the laser cutter was solved and we are now able to proceed with the cutting.
Gilbert got the working example for the top half.</td></tr>
<tr><td>Main business:</td><td colspan="2"></td><td>Action</td></tr>
<tr><td></td><td colspan="3">Iain laser cut the necessary components, but some of them didn’t have the right dimensions so had to be cut again.
Gilbert started to embed some of the electronic components onto the laser cut.
There were some issues with powering the colour sensor: it works fine when it’s powered by the laptop, but it doesn’t work with the battery. Gregor and Gilbert tried to fix it.

<u>For next meeting:</u></td></tr>
</table>

	Gilbert will bring a solar power battery and we'll see if the colour sensor works with that. Assemble the box.

25/04/2023

APPENDIX D FULL CODE

TOP HALF ARDUINO CODE

```
#include <Servo.h>
#include <LiquidCrystal_I2C.h>
#include <SoftwareSerial.h>

#define lockServoPin 5

#define buttonInputPin1 10
#define buttonInputPin2 11
#define buttonInputPin3 4
#define buttonInputPin4 3

// defines the initial button state
int buttonState1 = 0;
int buttonState2 = 0;
int buttonState3 = 0;
int buttonState4 = 0;

Servo lockServo;

LiquidCrystal_I2C lcd(0x27, 16, 2);

// serial communication definition
SoftwareSerial link(7, 2);

String colours[3] = {"red", "blue", "yellow"};
String currentColour = colours[0];
String selectedColour;
// initial colour id (red)
int colourID = 1;

// initial digitCode and setCode
int digitCode[3] = {0, 0, 0};
int setCode[3] = {0, 0, 0};


int trigPin = 8;    //Trig - green Jumper
int echoPin = 9;    //Echo - yellow Jumper
long duration, cm = 0;

/*
*  state index table
* 0 - colour selection state
* 1 - 3 didgit code state
* 2 - locking and checking for package
*   3 - unlocking stage
*/
```

Figure 40 Tops Half Code Part 1

```
// initial state
int currentState = 0;



void setup()
{
  // initialises the servo to be on digital pin 5
  lockServo.attach(5);

  // initialises the lcd screen
  lcd.init();
  lcd.backlight();

  // sets button pins to be input pins
  pinMode(buttonInputPin1, INPUT);
  pinMode(buttonInputPin2, INPUT);
  pinMode(buttonInputPin3, INPUT);
  pinMode(buttonInputPin4, INPUT);

  // ultrasonic distance sensor pins
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);

  // digital pin 2 is used for serial communication with other arduino boards
  pinMode(2, OUTPUT);
  Serial.begin(9600);
  link.begin(9600);

  // sets the servo so start in the unlocked position
  unlock();
}

// gets the current button state of all the buttons
void readButtonState(){

  buttonState1 = digitalRead(buttonInputPin1);
  buttonState2 = digitalRead(buttonInputPin2);
  buttonState3 = digitalRead(buttonInputPin3);
  buttonState4 = digitalRead(buttonInputPin4);

}
```

Figure 41 Tops Half Code Part 2

```
// moves the servo to the locked position
void lock(){

  lockServo.write(90);

}

// moves the servo to the unlocked position
void unlock(){

  lockServo.write(55);

}

// changes the selected colour based on the current button states
// changes currentState to 1 if the button4 is being pressed
// returns the selected colour
String selectColour(){

  if(buttonState1 == HIGH ){

    currentColour = colours[0];
    colourID = 0;

  }

  if(buttonState2 == HIGH ){

    currentColour = colours[1];
    colourID = 2;

  }

  if(buttonState3 == HIGH ){

    currentColour = colours[2];
    colourID = 1;
  }

  if(buttonState4 == HIGH ){
    // move to the next stage
  currentState = 1;
  }

  return currentColour;
}
```

Figure 42 Tops Half Code Part 3

```
// changes the 3 ddigit code based on the current button states
// changes currentState to 1 if the button4 is being pressed
// returns the new selected colour
void digitCodeSelection() {

  if(buttonState1 == HIGH ){

    digitCode[2]++;

    if (digitCode[2] >= 10){
      digitCode[2] = 0;
    }

  }

  if(buttonState2 == HIGH ){

    digitCode[1]++;

    if (digitCode[1] >= 10){
      digitCode[1] = 0;
    }
  }

  if(buttonState3 == HIGH ){

    digitCode[0]++;

    if (digitCode[0] >= 10){
      digitCode[0] = 0;
    }

  }

  if(buttonState4 == HIGH ){
    if (currentState == 1){
    currentState = 2;

      setCode[0] = digitCode[0];
        setCode[1] = digitCode[1];
        setCode[2] = digitCode[2];

      digitCode[0] = 0;
        digitCode[1] = 0;
        digitCode[2] = 0;
    }
  }
```

Figure 43 Tops Half Code Part 4

```
// compare each of values in setCode[3] to the corresponding values in inputCode[3]
// if the values stored in both arrays are the same true is returned otherwise false is returned
boolean checkDigitCode(int setCode[3], int inputCode[3]){

  if (setCode[0] == inputCode[0] && setCode[1] == inputCode[1] && setCode[2] == inputCode[2]){
    return true;
  }
  return false;
}

// uses data from an ultrasonic distance sensor to determine if a package is in the cabinet
boolean checkPackageInCase(){

  /*
  * Code excerpt written by Rui Santos, http://randomnerdtutorials.com
  * Complete Guide for Ultrasonic Sensor HC-SR04
  */

  digitalWrite(trigPin, LOW);
  delayMicroseconds(5);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);

  // Read the signal from the sensor: a HIGH pulse whose
  // duration is the time (in microseconds) from the sending
  // of the ping to the reception of its echo off of an object.
  pinMode(echoPin, INPUT);
  duration = pulseIn(echoPin, HIGH);

  // convert the time into a distance
  cm = (duration/2) / 29.1;

  /*
  * End of code excerpt written by Rui Santos
  */

  if(cm <= 12){
    return true;
  }

  return false;
}
```

Figure 44 Tops Half Code Part 5

```
// if a package is detected as being in the Case the lid is locked after a 10 second delay
// returns true if the lid has been locked otherwise returns false
boolean lockPackageInCase(){

  if (checkPackageInCase()){
    lcd.clear();
    lcd.print("Closing...");
    delay(10000);
    lock();
    return true;
  }

  return false;
}



void loop()
{

  // gets the current bottons pressed
  readButtonState();

  // clears the LCD screen to be written to
  lcd.clear();
  lcd.setCursor(0, 0);

  // code for transfering the colourID to the lower half has been commented out
  // due to the serial communication not working
  //byte msg = colourID;
  //link.write(msg);

    switch (currentState){

      // colour selection case
      case 0:

      selectedColour = selectColour();
        lcd.print(selectedColour);

        delay(200);
      break;
```

Figure 45 Tops Half Code Part 6

```
        // code selection case
        case 1:

          digitCodeSelection();

          lcd.print(digitCode[0]);
          lcd.print(digitCode[1]);
          lcd.print(digitCode[2]);

          delay(200);
          break;

        // cabinet locking case
        case 2:
          // moves to the next stage if the package has been locked in the cabinet
          if (lockPackageInCase()){
            currentState = 3;
          }
          // otherwise displays the pacage message
          else{
            lcd.setCursor(0, 0);
            lcd.print("place package");
            lcd.setCursor(0, 1);
            lcd.print("And Close Lid");

          }
          delay(200);

          break;

        // unlocking case
        case 3:
          // select a 3 digit code
      digitCodeSelection();

          lcd.print(digitCode[0]);
          lcd.print(digitCode[1]);
          lcd.print(digitCode[2]);

          // compare the code to the previously set code
          if (checkDigitCode(setCode, digitCode) && buttonState4 == LOW){
            unlock();
            lcd.clear();
            lcd.print("unlocked");
            delay(10000);
            currentState = 0;
          }

          delay(200);
          break;

    }
}
```

Figure 46 Tops Half Code Part 7

BOTTOM HALF ARDUINO CODE

```
#include <SoftwareSerial.h>
#include <Adafruit_TCS34725.h>

#define rxPin 2

#define HBRIDGE_LEFT_PIN_A 5
#define HBRIDGE_LEFT_PIN_B 6

#define HBRIDGE_RIGHT_PIN_A 10
#define HBRIDGE_RIGHT_PIN_B 11

#define trigPin 7
#define echoPin 8

#define buzzerPin 12

Adafruit_TCS34725 tcs = Adafruit_TCS34

long duration = 0.0, cm;

float motorSpeedLeft = 200;
float motorSpeedRight = 200;

int maxSpeed = 160;

const int irPins[3] = {A0, A1, A3};

int irSensorDigital[3] = {0, 0, 0};

int difThreshold = 500;

int irSensors = B000;

int error = 0;

int errorLast = 0;

// SoftwareSerial setup
SoftwareSerial ss(2,3);
char cString[20];
byte chPos = 0;
byte ch = 0;

int destinationColorID = 0;
int colorID = -1;
int colourSensorCount = 0;
```

Figure 47 Tops Bottom Code Part 1

```
int cmod = 0;

bool pressed = false;

void setup()
{
  Serial.begin(9600);
  pinMode(HBRIDGE_LEFT_PIN_A,OUTPUT); //Control Direction of motor.
  pinMode(HBRIDGE_LEFT_PIN_B,OUTPUT); //Control Direction of motor.
  digitalWrite(HBRIDGE_LEFT_PIN_A,LOW); // Start with motors stopped
  digitalWrite(HBRIDGE_LEFT_PIN_B,LOW);

  pinMode(HBRIDGE_RIGHT_PIN_A,OUTPUT);
  pinMode(HBRIDGE_RIGHT_PIN_B,OUTPUT);
  digitalWrite(HBRIDGE_RIGHT_PIN_A,LOW);
  digitalWrite(HBRIDGE_RIGHT_PIN_B,LOW);

  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);

  pinMode(13,INPUT);

  tcs.begin();

  ss.begin(9600);
  pinMode(rxPin,INPUT);
}

// code from course notes
void scan() {

  irSensors = B000;

  for (int i = 0; i < 3; i++) {
      int sensorValue = analogRead(irPins[i]);
      if (sensorValue >= difThreshold) {
        irSensorDigital[i] = 1;
      }

    else {
      irSensorDigital[i] = 0;
    }

    int b = 2-i;
    irSensors = irSensors + (irSensorDigital[i]<<b);
```

Figure 48 Tops Bottom Code Part 2

```
    }
}

// code from course notes
// error values have been calibrated for the robot
void updateDirection(int motorSpeedleft, int motorSpeedright) {

  errorLast = error;
  switch (irSensors) {

    case B000: // no sensor detects the line
       if (errorLast < 0) { error = -125;}
       else if (errorLast > 0) {error = 125;}
       break;

     case B100: // left sensor on the line
       error = 70;
       break;

     case B110:
       error = -35;
       break;

     case B010:
       error = 0;
       break;

     case B011:
       error = 35;
       break;

     case B001: // right sensor on the line
       error = -70;
       break;

     case B111:
       error = 0;
       break;

     default:
       error = errorLast;
  }

    if (error > 0) {
      motorSpeedLeft = maxSpeed;
```

Figure 49 Tops Bottom Code Part 3

```
      motorSpeedRight = maxSpeed - error;
    }

    else if (error < 0) {
      motorSpeedLeft = maxSpeed + error;
      motorSpeedRight = maxSpeed;
    }

}

// code from course notes
void set_motor_pwm(int pwm, int IN1_PIN, int IN2_PIN)
{
  if (pwm < 0) {  // reverse speeds
    analogWrite(IN1_PIN, -pwm);
    digitalWrite(IN2_PIN, LOW);

  } else { // stop or forward
    digitalWrite(IN1_PIN, LOW);
    analogWrite(IN2_PIN, pwm);
  }
}

// code from course notes
void set_motor_left(int speed)
{
  // Sets the direction of the motor (See Notes about H-bridges)
  set_motor_pwm(speed,HBRIDGE_LEFT_PIN_B,HBRIDGE_LEFT_PIN_A);
}

// code from course notes
void set_motor_right(int speed)
{
    // Sets the direction of the motor (See Notes about H-bridges)
  set_motor_pwm(speed,HBRIDGE_RIGHT_PIN_B,HBRIDGE_RIGHT_PIN_A);
}
```

Figure 50 Tops Bottom Code Part 4

```
void echo(){

  /*
  * Code excerpt written by Rui Santos, http://randomnerdtutorials.com
  *  Complete Guide for Ultrasonic Sensor HC-SR04
  */

  // The sensor is triggered by a HIGH pulse of 10 or more microseconds.
  // Give a short LOW pulse beforehand to ensure a clean HIGH pulse:
  digitalWrite(trigPin, LOW);
  delayMicroseconds(5);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);

  // Read the signal from the sensor: a HIGH pulse whose
  // duration is the time (in microseconds) from the sending
  // of the ping to the reception of its echo off of an object.
  pinMode(echoPin, INPUT);
  duration = pulseIn(echoPin, HIGH,10000);

  // convert the time into a distance
  cm = (duration/2) / 29.1;

  /*
  * End of code excerpt written by Rui Santos
  */

  // if the ultrasonic distance sensor fails to determine a distance cm is set to a very high value
  if (duration == 0){
    cm = 1000000;
  }
}
```

Figure 51 Tops Bottom Code Part 5

```
// colourDistance determines how similar a set of RGB values is to a defined color[]
float colorDistance(int r,int  g,int b,int color[]){
  float rd = color[0] - r;
  float gd = color[1] - g;
  float bd = color[2] - b;
  return sqrt(rd*rd + gd*gd + bd*bd);
}
```

Figure 52 Tops Bottom Code Part 6

```
// returns the colour ID of the colour that most closely matches the colour beneath the sensor
int detectColour(){
  float r, g, b;
  // gets RGB values from the sensor
  tcs.getRGB(&r, &g, &b);

  //Serial.print(r);
  //Serial.print(" ");
  //Serial.print(g);
  //Serial.print(" ");
  //Serial.println(b);

  // defined colours used in colorDistance()
  int blue[] = {78,78,117};
  int yellow[] = {120,85,47};
  int red[] = {150,60,60};
  int green[] = {71,115,78};
  int tape[] = {90,90,75};
  int ground[] = {90,85,80};

  // determining the distance to each defined colour
  float red_dist = colorDistance(r,g,b,red);
  float yellow_dist = colorDistance(r,g,b,yellow);
  float blue_dist = colorDistance(r,g,b,blue);
  float green_dist = colorDistance(r,g,b,green);
  float tape_dist = colorDistance(r,g,b,ground);
  float ground_dist = colorDistance(r,g,b,ground);

  // finds the colours which is matches closest
  float mn = min(min(tape_dist, green_dist),min(min(red_dist,yellow_dist),min(blue_dist,ground_dist)));

  // returns the colourID of the closest colour
  if (mn == red_dist){
    return 0;
  }
  else if (mn == yellow_dist){
    return 1;
  }
  else if (mn == blue_dist){
    return 2;
  }
  else if (mn == green_dist){
    return 3;
  }
  else if (mn == ground_dist or mn == tape_dist){
    return -1;
  }
  return -1;
}
```

Figure 53 Tops Bottom Code Part 7

```
void loop()
{
  // checks the IR sensor values and updates the speeds on the motors to follow the line
  scan();
  updateDirection(motorSpeedLeft, motorSpeedRight);

  // uses the echo() function to find the distance of obstacles
  echo();
  // halts motor movement if the obstacle is too close
  if (cm <= 30){
      set_motor_right(0);
      set_motor_left(0);

      // plays buzzer tones to notify people of the obstacle blocking the path
      tone(buzzerPin, 1000);
      delay(250);
      tone(buzzerPin, 500);
      delay(250);
      tone(buzzerPin, 2000);
      delay(250);
      tone(buzzerPin, 1500);
      delay(250);
      noTone(buzzerPin);
      delay(1000);

  } else{

      set_motor_left(motorSpeedLeft);
      set_motor_right(motorSpeedRight);

  }
```

Figure 54 Tops Bottom Code Part 8

```
  // gets the ID of the colour of the surface beneath the colour sensor
  colorID = detectColour();

  // code for cycling through the colours for destinationColorID
  // this is used since serial communication was not working properly
  if (digitalRead(13) == HIGH and pressed == false){
    tone(buzzerPin,500);
    delay(50);
    noTone(buzzerPin);
    pressed = true;
    cmod++;
  }
  else if (digitalRead(13) == LOW){
    pressed = false;
  }

  destinationColorID = cmod % 3;

// code has been commented out and replaced due to serial communication not working properly
//  if(ss.available())
//  {
//    int serialInput = ss.read();
//    serialInput = 0;
//    switch(serialInput){
//      case 0: //change color
//          destinationColorID = serialInput;
//      case 1: //change color
//          destinationColorID = serialInput;
//      case 2: //change color
//          destinationColorID = serialInput;
////      case 3:
////          tone(buzzerPin,2500);
////          delay(50);
////          noTone(buzzerPin);
////          set_motor_left(0);
////          set_motor_right(0);
////          delay(1000);
//    }
//  }
```

Figure 55 Tops Bottom Code Part 9

```
  // turns left when the destination colour is detected
  if (colorID == destinationColorID){
    set_motor_left(200);
    set_motor_right(100);
    tone(buzzerPin,700);
    delay(50);
    noTone(buzzerPin);
    delay(1000);
  }
  else{
    maxSpeed = 160;
  }

  // if Green is detected the robot has reached the end point
  // stops motor movement and plays some buzzers tones
  if (colorID == 3){
    set_motor_right(0);
    set_motor_left(0);
    tone(buzzerPin,250);
    delay(50);
    noTone(buzzerPin);
    delay(50);
    tone(buzzerPin,250);
    delay(50);
    noTone(buzzerPin);
    delay(1000);
  }
}
```

Figure 56 Tops Bottom Code Part 10